

Table 2 Experimental results on the pure copper surface under different steady state

No.	T_1 (°C)	T_2 (°C)	T_3 (°C)	T_{sat} (°C)	q'' (kW/m ²)	T_{wall} (°C)	ΔT_{wall} (°C)	h (kW/(m ² ·K))
1	107.55	107.99	109.24	101.67	41.18	105.97	4.30	9.59
2	112.10	114.05	116.58	101.74	109.23	107.90	6.16	17.73
3	115.74	119.42	123.04	101.77	177.65	108.91	7.14	24.88
4	117.72	122.37	126.87	101.73	222.88	109.14	7.42	30.05
5	120.64	126.66	132.33	101.66	284.65	109.69	8.02	35.47
6	123.57	131.00	137.91	101.68	349.14	110.14	8.46	41.25
7	126.39	135.13	143.20	101.76	409.31	110.65	8.89	46.06
8	128.78	138.77	147.90	101.64	465.43	110.87	9.23	50.40
9	131.58	142.91	153.19	101.68	525.98	111.35	9.68	54.36
10	134.20	146.74	158.20	101.66	584.17	111.73	10.07	58.01
11	136.80	150.56	163.03	101.66	638.52	112.24	10.58	60.35
12	139.66	154.81	168.40	101.69	699.51	112.76	11.07	63.17
13	142.74	159.41	174.25	101.61	766.94	113.24	11.64	65.91
14	145.24	163.07	178.81	101.71	817.13	113.81	12.10	67.50
15	147.88	166.94	183.74	101.70	872.84	114.31	12.62	69.19
16	150.48	170.76	188.55	101.71	926.70	114.84	13.13	70.60
17	153.47	175.13	193.99	101.72	986.35	115.53	13.81	71.41
18	156.34	179.26	199.17	101.69	1042.64	116.24	14.54	71.69
19	158.88	182.90	203.80	101.70	1093.28	116.83	15.13	72.25

rate of increment somewhat decreases at the high heat flux. The HTC reaches the peak when the CHF phenomenon occurs.

All of the steady-state condition data are shown in Table 2 and the corresponding visualization experimental results are shown in Fig. 3. From visual inspection through high-speed CCD, at the beginning of the experiment, the bubbles are small and independent (Fig. 3(a)). With the rising of the heat flux, the small bubbles gather together and grow larger on the copper surface (Figs. 3(b) and 3(c)). The larger the heat flux rose, the growth and enlargement of localized hot/dry regions are easier to form large vapor slugs (Figs. 3(d) and 3(e)). The vapor slugs and bubbles will vigorously coalesce even further when the heat flux keeps increasing, until the formation of a vapor blanket that covers the boiling surface, which means a boiling regime transition (Fig. 3(f)).

4 Conclusions

In this work, the recent saturated pool boiling experimental results were introduced, which were conducted at atmospheric pressure using the pure copper as heating surface and the distilled water as working fluid. The HTC is 72.25 kW/(m²·K) and the CHF is 1093.28 kW/m², which were obtained by the specially designed pool boiling experimental

device. Furthermore, the visualization experimental results of pool boiling, such as typical nucleate boiling and typical film boiling, have been captured by the high-speed camera in order to provide references for future boiling mechanism research.

The experimental results have shown the skillfully master of pool boiling experimental technology. In future work, the proposed approach is easily scalable to various material surfaces or different working fluids, for example, the graphene coating surface or nanofluid. The pool boiling experiment would be the benchmark data in the future experiments. Furthermore, the computer simulations would be validated by the experimental results.

In conclusion, it is believed that the boiling experiment in the current work is of great importance for the enhancement of efficiency and safety in the energy industry, particularly demonstrating the feasibility of developing a high-performance and durable heat exchange surface.

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